

Problem 4 — Full Walkthrough

Linewise local minima

Setup: Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be differentiable, and suppose x^* is a local minimum of f along every line through x^* — that is, for every direction $d \in \mathbb{R}^n$, the function $g(\alpha) = f(x^* + \alpha d)$ has a local minimum at $\alpha = 0$.

Part (a) — Show $\nabla f(x^*) = 0$

Step 1 — Fix an arbitrary direction

Fix any $d \in \mathbb{R}^n$ (this is given to us by hypothesis — the property holds for *every* d , so we're free to reason about one arbitrary, unspecified d and conclude something that then holds for all of them). Define

$$g(\alpha) := f(x^* + \alpha d), \quad \alpha \in \mathbb{R}.$$

Step 2 — Differentiate g via the chain rule

Write $g(\alpha) = f(h(\alpha))$ where $h(\alpha) := x^* + \alpha d$. Since h is affine, $h'(\alpha) = d$ for every α . By the (multivariable) chain rule,

$$g'(\alpha) = \nabla f(h(\alpha))^T h'(\alpha) = \nabla f(x^* + \alpha d)^T d.$$

In particular, at $\alpha = 0$:

$$g'(0) = \nabla f(x^*)^T d.$$

Step 3 — Use the local-min hypothesis on g

By assumption, g has a local minimum at $\alpha = 0$. Since $\alpha = 0$ is an interior point of g 's domain ($\mathbb{R}$) and g is differentiable there, the standard first-order necessary condition for a one-variable local minimum applies: $g'(0) = 0$. (If $g'(0)$ were nonzero, moving α slightly in the appropriate direction would strictly decrease g , contradicting a local minimum at 0.)

Step 4 — Combine Steps 2 and 3

$$\nabla f(x^*)^T d = g'(0) = 0.$$

Since d was an arbitrary direction (Step 1), this holds for *every* $d \in \mathbb{R}^n$:

$$\nabla f(x^*)^T d = 0, \quad \forall d \in \mathbb{R}^n.$$

Step 5 — Choose the one direction that finishes the proof

This is the key trick: since the equation above holds for literally every d , we are free to plug in whichever d is most convenient. Choose

$$d := \nabla f(x^*).$$

Substituting:

$$\nabla f(x^*)^T \nabla f(x^*) = 0 \implies \|\nabla f(x^*)\|^2 = 0.$$

Step 6 — Conclude

Write out the squared norm: $\|\nabla f(x^*)\|^2 = (\partial f/\partial x_1)^2 + \cdots + (\partial f/\partial x_n)^2$, a sum of squares. A sum of nonnegative terms equals zero only if every term is individually zero. So each partial derivative of f at x^* is zero, i.e.

$$\nabla f(x^*) = 0. \quad \blacksquare$$

Part (b) — An example where x^* is not actually a local minimum

Step 1 — The example

Let $0 < p < q$, and define

$$f(y, z) = (z - py^2)(z - qy^2).$$

We'll check the point $x^* = (0, 0)$: $f(0, 0) = (0 - 0)(0 - 0) = 0$.

Step 2 — Verify $(0, 0)$ is a local minimum along every line through it

Take an arbitrary direction $d = (a, b) \neq (0, 0)$, and parametrize the line as $(y, z) = (\alpha a, \alpha b)$:

$$g(\alpha) := f(\alpha a, \alpha b) = (\alpha b - p\alpha^2 a^2)(\alpha b - q\alpha^2 a^2).$$

Case $a = 0$ (so $b \neq 0$):

$$g(\alpha) = (\alpha b - 0)(\alpha b - 0) = (\alpha b)^2 \geq 0,$$

with equality only at $\alpha = 0$. So g has a strict global (hence local) minimum at $\alpha = 0$.

Case $a \neq 0$, $b \neq 0$: factor out α^2 :

$$g(\alpha) = \alpha^2 [(b - p\alpha a^2)(b - q\alpha a^2)].$$

As $\alpha \rightarrow 0$, the bracketed factor is continuous and approaches $b \cdot b = b^2 > 0$. So there is some neighborhood of $\alpha = 0$ where the bracket stays strictly positive. On that neighborhood, $g(\alpha) = \alpha^2 \times (\text{positive number}) \geq 0$, with equality only at $\alpha = 0$ (since $\alpha^2 = 0$ only there). So g has a strict local minimum at $\alpha = 0$.

Case $a \neq 0$, $b = 0$:

$$g(\alpha) = (0 - p\alpha^2 a^2)(0 - q\alpha^2 a^2) = (p\alpha^2 a^2)(q\alpha^2 a^2) = pq\alpha^4 a^4 \geq 0,$$

with equality only at $\alpha = 0$ (since $p, q, a^4 > 0$). Local minimum at $\alpha = 0$.

Every possible direction $(a, b) \neq (0, 0)$ falls into one of these three cases, and in each one $g(\alpha) \geq g(0) = 0$ near $\alpha = 0$. So $(0, 0)$ is a local minimum of f along every single line through it.

Step 3 — Show $(0,0)$ is *not* actually a local minimum of f

The hint suggests looking off the straight lines entirely — along the curved path $z = my^2$ for some fixed m with $p < m < q$. Substitute $z = my^2$ into f :

$$f(y, my^2) = (my^2 - py^2)(my^2 - qy^2) = y^2(m - p) \cdot y^2(m - q) = y^4(m - p)(m - q).$$

Since $p < m < q$: $(m - p) > 0$ and $(m - q) < 0$, so their product is negative. And $y^4 \geq 0$, strictly positive whenever $y \neq 0$. So:

$$f(y, my^2) = y^4 \cdot \underbrace{(m - p)(m - q)}_{<0} < 0 \quad \text{for every } y \neq 0.$$

Meanwhile $f(0,0) = 0$. As $y \rightarrow 0$, the points (y, my^2) approach $(0,0)$, and along this entire path f stays strictly negative — meaning **every neighborhood of $(0,0)$, no matter how small, contains points where f is strictly less than $f(0,0)$** . That directly violates the definition of a local minimum (which requires $f(x) \geq f(x^*)$ for *all* nearby x , not just along straight lines). **So $(0,0)$ is not a local minimum of f . ■**

Why this doesn't contradict part (a): a consistency check

Part (a) says a linewise local minimum must have zero gradient. It does *not* say a linewise local minimum must be an actual local minimum — and this example is precisely why that distinction matters. As a sanity check, let's confirm $\nabla f(0,0) = 0$ really does hold here, directly from the formula (as it must, by part (a), since we just verified $(0,0)$ satisfies the linewise-minimum hypothesis).

Let $u = z - py^2$, $w = z - qy^2$, so $f = uw$. By the product rule:

$$\frac{\partial f}{\partial z} = w + u = (z - qy^2) + (z - py^2) = 2z - (p + q)y^2,$$

$$\frac{\partial f}{\partial y} = -2py \cdot w - 2qy \cdot u = -2py(z - qy^2) - 2qy(z - py^2) = -2(p + q)yz + 4pqy^3.$$

At $(y, z) = (0,0)$:

$$\left. \frac{\partial f}{\partial z} \right|_{(0,0)} = 2(0) - (p + q)(0)^2 = 0, \quad \left. \frac{\partial f}{\partial y} \right|_{(0,0)} = -2(p + q)(0)(0) + 4pq(0)^3 = 0.$$

So $\nabla f(0,0) = (0,0)$, confirming directly what part (a) guarantees. The point is a genuine stationary point — flat in every direction — yet still not a local minimum, because the function dips below its value at $(0,0)$ along a curved path that no straight line happens to detect. This is exactly why second-order conditions (or, as here, direct inspection along curves) are sometimes needed beyond the gradient test.